

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I V.Raj kumar(192525208)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Raj kumar

Criteria	Understanding of Scenario (1)	Identification of Risks / Issues (1)	Application of Concepts (1.25)	Decision Justification (1)	Clarity (0.75)	TOTAL (5)
Weightage	20%	20%	25%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Scenario-Based Questions

1. A financial institution plans to deploy a private cloud with strict compliance requirements. Recommend an appropriate cloud datacenter architecture and explain how virtualization and isolation mechanisms can ensure security and regulatory compliance.
2. A SaaS provider experiences inconsistent application performance due to noisy neighbor effects in a virtualized environment. Analyze the root cause and propose resource allocation and hypervisor-level solutions.
3. An organization wants to migrate legacy monolithic applications into a Service-Oriented Architecture (SOA) in the cloud. Explain the migration strategy and how SOA improves scalability and maintainability.
4. A government agency is implementing identity federation across multiple departments using cloud services. Explain how identity management, authentication, and privacy policies should be designed to ensure secure access.
5. A cloud provider wants to implement real-time user presence detection for collaborative applications. Discuss how presence services can be integrated into cloud architecture while maintaining user privacy and system scalability.

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COURSE CODE:CSA1504

COURSENAME:CLOUD COMPUTING AND BIG DATA ANALYSIS

Q1. Private Cloud with Compliance Requirements

Answer:

A financial institution should use a **Private Cloud Datacenter Architecture** because it provides high security, full control, and regulatory compliance. The datacenter should include secure servers, storage, networking, firewalls, and virtualization software.

Virtualization and Isolation Mechanisms:

- Use a **Type-1 Hypervisor** (VMware ESXi, Hyper-V, Xen) for better security.
- Create separate Virtual Machines (VMs) for different departments.
- Apply network isolation using VLANs and firewalls.
- Encrypt stored and transmitted data.
- Enable role-based access control (RBAC) and multi-factor authentication (MFA).
- Maintain audit logs and regular security monitoring to meet compliance standards.

Conclusion:

A private cloud with virtualization and isolation ensures data confidentiality, prevents unauthorized access, and satisfies regulatory requirements.

Q2. Noisy Neighbor Problem in SaaS

Answer:

The inconsistent performance is caused by the **Noisy Neighbor Effect**, where multiple virtual machines share the same physical resources. If one VM consumes excessive CPU, memory, or storage, other VMs experience reduced performance.

Solutions:

- Allocate dedicated CPU, memory, and storage resources.
- Use resource limits and reservations.
- Enable hypervisor scheduling to balance workloads.
- Perform live migration of overloaded VMs.
- Monitor resource usage continuously.
- Implement Quality of Service (QoS) policies.

Conclusion:

Proper resource allocation and hypervisor management eliminate noisy neighbor issues and improve application performance.

Q3. Migrating Legacy Applications to SOA

Answer:

The organization should gradually migrate the monolithic application into **Service-Oriented Architecture (SOA)** by dividing it into independent services.

Migration Strategy:

1. Analyze the existing application.
2. Identify business functions.
3. Convert each function into a reusable service.
4. Develop APIs for communication.
5. Deploy services on the cloud.
6. Test and monitor all services.

Advantages of SOA:

- Better scalability.
- Easier maintenance.
- Faster deployment.
- Reusable services.
- Improved fault isolation.

Conclusion:

SOA increases flexibility, scalability, and maintainability while reducing system complexity.

Q4. Identity Federation in Cloud

Answer:

Identity federation allows users to access multiple cloud services with a single login.

Secure Identity Management:

- Implement Single Sign-On (SSO).
- Use Identity Providers (IdP) and Service Providers (SP).
- Support SAML, OAuth, or OpenID Connect.
- Enable Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Encrypt identity information.

- Maintain audit logs and privacy policies.

Conclusion:

Identity federation provides secure, convenient, and centralized authentication while protecting user privacy.

Q5. Presence Services in Cloud**Answer:**

Presence services detect and display whether users are online, offline, busy, or available in real time.

Integration in Cloud Architecture:

- Use a centralized presence server.
- Update user status in real time.
- Synchronize data across devices.
- Use scalable cloud infrastructure with load balancing.
- Encrypt presence information.
- Allow users to control privacy settings.
- Restrict access using authentication and authorization.

Conclusion:

Presence services improve collaboration while maintaining user privacy, security, and cloud scalability.